Physical Chemistry Cume
April 3, 2015

1)

For the following reaction to produce phosgene, Cl_2CO, from Cl_2 and CO:

$$Cl_{2(g)} + CO_{(g)} \xrightarrow{k_{obs}} Cl_2CO_{(g)}$$

Assuming that the reaction proceeds via the following mechanism

$$Cl_{2(g)} + M_{(g)} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2Cl_{(g)} + M_{(g)} \qquad \text{(fast equilibrium)}$$

$$Cl_{(g)} + CO_{(g)} + \mathrm{M}_{(g)} \underset{k_{-2}}{\overset{k_2}{\rightleftharpoons}} ClCO_{(g)} + M_{(g)} \qquad \text{(fast equilibrium)}$$

$$ClCO_{(g)} + Cl_{2(g)} \xrightarrow{k_3} Cl_2CO_{(g)} + Cl_{(g)} \qquad \text{(slow)}$$

Where M is any gas molecule in the container

a) Find the rate law for the formation of phosgene, $Cl_2CO_{(g)}$

b) express k_{obs} in terms of the rate constants for the individual steps of the reaction

c) What is the role of M? Would the rate would show a pressure dependence on $M_{(g)}$? explain.

Question 2 on next page ...

2)

HF can be formed from H_2 and F_2

$$H_{2(g)} + F_{2(g)} \xrightarrow{k_{obs}} 2HF_{(g)}$$

The mechanism for this reaction involves the elementary steps:

$$F_{2(g)} + \mathrm{M}_{(g)} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2\mathrm{F}_{(g)} + M_{(g)} \qquad \Delta_r H^0_{298K} = +159 kJ/mol$$

$$F_{(g)} + H_{2(g)} \underset{k_{-2}}{\overset{k_2}{\rightleftharpoons}} \mathrm{HF}_{(g)} + H_{(g)} \qquad \Delta_r H^0_{298K} = -134 kJ/mol$$

$$H_{(g)} + F_{2(g)} \xrightarrow{k_3} \mathrm{HF}_{(g)} + F_{(g)} \qquad \Delta_r H^0_{298K} = -411 kJ/mol$$

Where M is any gas molecule in the container

a) Why is the reaction

$$H_{2(g)} + \mathrm{M}_{(g)} \rightleftharpoons 2H_{(g)} + M_{(g)}$$

not included in the mechanism even though it produces the reactant H, which is used in step 3?

b) Derive the rate law for the formation of HF assuming a steady state condition applies to the intermediate species, expressing k_{obs} in terms of the rate constants for the individual steps.

c) Rationalize whether you think the steady-state assumption is valid or not.